BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/hw2-alex_matt`
   Installed InlineStrings         v1.4.5
   Installed Libmount_jll          v2.41.1+0
   Installed JpegTurbo_jll         v3.1.3+0
   Installed InvertedIndices        v1.3.1
   Installed Accessors           v0.1.42
   Installed Roots             v2.2.10
   Installed Preferences          v1.5.0
   Installed DataFrames          v1.8.0
   Installed Fontconfig_jll         v2.17.1+0
   Installed WorkerUtilities        v1.6.1
   Installed SentinelArrays         v1.4.8
   Installed WeakRefStrings         v1.4.2
   Installed StatsBase           v0.34.6
   Installed DataStructures         v0.19.1
   Installed Expat_jll           v2.7.1+0
   Installed ConstructionBase        v1.6.0
   Installed PooledArrays          v1.4.3
   Installed Ghostscript_jll        v9.55.1+0
   Installed TableTraits          v1.0.1
   Installed DataValueInterfaces       v1.0.0
   Installed CompositionsBase        v0.1.2
   Installed CommonSolve          v0.2.4
   Installed Plots             v1.40.19
   Installed PrettyTables          v3.0.8
   Installed Libuuid_jll          v2.41.1+0
   Installed InverseFunctions        v0.1.17
   Installed OpenSSL_jll          v3.5.2+0
   Installed Latexify            v0.16.10
   Installed Glib_jll            v2.86.0+0
   Installed FilePathsBase          v0.9.24
   Installed IteratorInterfaceExtensions  v1.0.0
   Installed Crayons             v4.1.1
   Installed CSV              v0.10.15
   Installed Tables             v1.12.1
   Installed StringManipulation       v0.4.1
   Installed SortingAlgorithms       v1.2.2
Precompiling project...
    351.1 ms   IteratorInterfaceExtensions
    299.0 ms   CommonSolve
    298.9 ms   DataValueInterfaces
    330.1 ms   WorkerUtilities
    338.2 ms   InverseFunctions
```

```
 264.6 ms   CompositionsBase
 303.0 ms   InvertedIndices
 331.4 ms   ConstructionBase
 448.9 ms   InlineStrings
 391.2 ms   PooledArrays
 729.5 ms   Crayons
 523.6 ms   Preferences
 424.7 ms   TableTraits
 591.8 ms   FilePathsBase
 938.3 ms   SentinelArrays
 336.6 ms   InverseFunctions → InverseFunctionsDatesExt
 589.3 ms   InverseFunctions → InverseFunctionsTestExt
1349.1 ms   DataStructures
 390.9 ms   LogExpFunctions → LogExpFunctionsInverseFunctionsExt
 551.7 ms   Unitful → InverseFunctionsUnitfulExt
 443.2 ms   CompositionsBase → CompositionsBaseInverseFunctionsExt
 525.6 ms   ConstructionBase → ConstructionBaseLinearAlgebraExt
 475.4 ms   Unitful → ConstructionBaseUnitfulExt
 399.2 ms   PrecompileTools
 615.7 ms   JLLWrappers
 514.4 ms   FilePathsBase → FilePathsBaseMmapExt
 816.6 ms   Tables
 610.7 ms   SortingAlgorithms
 970.2 ms   FilePathsBase → FilePathsBaseTestExt
1195.0 ms   RecipesBase
1268.7 ms   StringManipulation
1780.8 ms   Accessors
 591.1 ms   OpenSSL_jll
 355.0 ms   Libmount_jll
 380.9 ms   Graphite2_jll
 370.1 ms   EpollShim_jll
 366.9 ms   LLVMOpenMP_jll
 371.2 ms   Bzip2_jll
 345.9 ms   Xorg_libICE_jll
 358.0 ms   Xorg_libXau_jll
 416.3 ms   libpng_jll
 406.7 ms   libfdk_aac_jll
 411.5 ms   LAME_jll
 391.3 ms   LERC_jll
 408.2 ms   fzf_jll
 454.2 ms   JpegTurbo_jll
 429.8 ms   XZ_jll
3030.2 ms   ColorSchemes
```

```
 423.0 ms    Ogg_jll
 365.8 ms    mtdev_jll
 350.9 ms    Xorg_libXdmcp_jll
 414.8 ms    x265_jll
 415.6 ms    x264_jll
 408.1 ms    libaom_jll
 408.8 ms    Zstd_jll
 401.1 ms    Expat_jll
 396.7 ms    LZO_jll
 368.1 ms    Xorg_xtrans_jll
 379.8 ms    Opus_jll
 361.1 ms    libevdev_jll
 359.7 ms    eudev_jll
 399.8 ms    Libffi_jll
 421.3 ms    Libiconv_jll
 371.6 ms    Libuuid_jll
 426.6 ms    FriBidi_jll
 709.7 ms    Accessors → LinearAlgebraExt
 498.8 ms    Accessors → TestExt
 562.4 ms    Accessors → UnitfulExt
1883.6 ms    StatsBase
 602.5 ms    Pixman_jll
 392.6 ms    Xorg_libSM_jll
 628.6 ms    FreeType2_jll
1852.1 ms    OpenSSL
 797.7 ms    JLFzf
7968.7 ms    Parsers
 380.5 ms    Xorg_libxcb_jll
1165.8 ms    Ghostscript_jll
 560.7 ms    libvorbis_jll
 353.5 ms    Dbus_jll
 355.5 ms    libinput_jll
 390.6 ms    Wayland_jll
 864.2 ms    Libtiff_jll
 652.1 ms    GettextRuntime_jll
1212.6 ms    Fontconfig_jll
2227.2 ms    Roots
1141.0 ms    JSON
 524.7 ms    InlineStrings → ParsersExt
 383.3 ms    Xorg_xcb_util_jll
 405.3 ms    Xorg_libX11_jll
 910.1 ms    Glib_jll
 669.6 ms    Roots → RootsUnitfulExt
```

```
 6753.9 ms   PlotUtils
 2521.6 ms   Latexify
  381.3 ms   Xorg_xcb_util_image_jll
  729.9 ms   WeakRefStrings
  364.6 ms   Xorg_xcb_util_keysyms_jll
  350.3 ms   Xorg_xcb_util_renderutil_jll
  361.9 ms   Xorg_xcb_util_wm_jll
  362.1 ms   Xorg_libXrender_jll
  371.1 ms   Xorg_libXext_jll
  367.4 ms   Xorg_libXfixes_jll
  366.9 ms   Xorg_libxkbfile_jll
 1019.5 ms   UnitfulLatexify
  697.1 ms   Latexify → SparseArraysExt
 2299.7 ms   PlotThemes
  486.8 ms   Xorg_xcb_util_cursor_jll
  427.0 ms   Xorg_libXinerama_jll
 3318.0 ms   RecipesPipeline
  406.7 ms   Xorg_libXrandr_jll
  439.3 ms   Libglvnd_jll
  442.9 ms   Xorg_libXcursor_jll
  443.1 ms   Xorg_libXi_jll
  446.3 ms   Xorg_xkbcomp_jll
 1969.0 ms   Cairo_jll
  538.7 ms   Xorg_xkeyboard_config_jll
  395.5 ms   xkbcommon_jll
  890.0 ms   HarfBuzz_jll
  468.9 ms   Vulkan_Loader_jll
12978.5 ms   HTTP
  806.1 ms   libass_jll
  820.7 ms   Pango_jll
  397.9 ms   libdecor_jll
  411.5 ms   GLFW_jll
 2369.0 ms   FFMPEG_jll
 3080.1 ms   Qt6Base_jll
  967.8 ms   FFMPEG
 1209.3 ms   Qt6ShaderTools_jll
 1281.0 ms   GR_jll
22790.3 ms   PrettyTables
10629.7 ms   CSV
 1258.4 ms   Qt6Declarative_jll
  282.7 ms   Qt6Wayland_jll
 2243.7 ms   GR
26068.1 ms   DataFrames
```

```
  1446.5 ms    Latexify → DataFramesExt
 34592.1 ms    Plots
  2804.6 ms    Plots → UnitfulExt
 137 dependencies successfully precompiled in 72 seconds. 82 already precompiled.
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in $^\circ$C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/$^\circ$C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproteorozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/$^\circ$C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

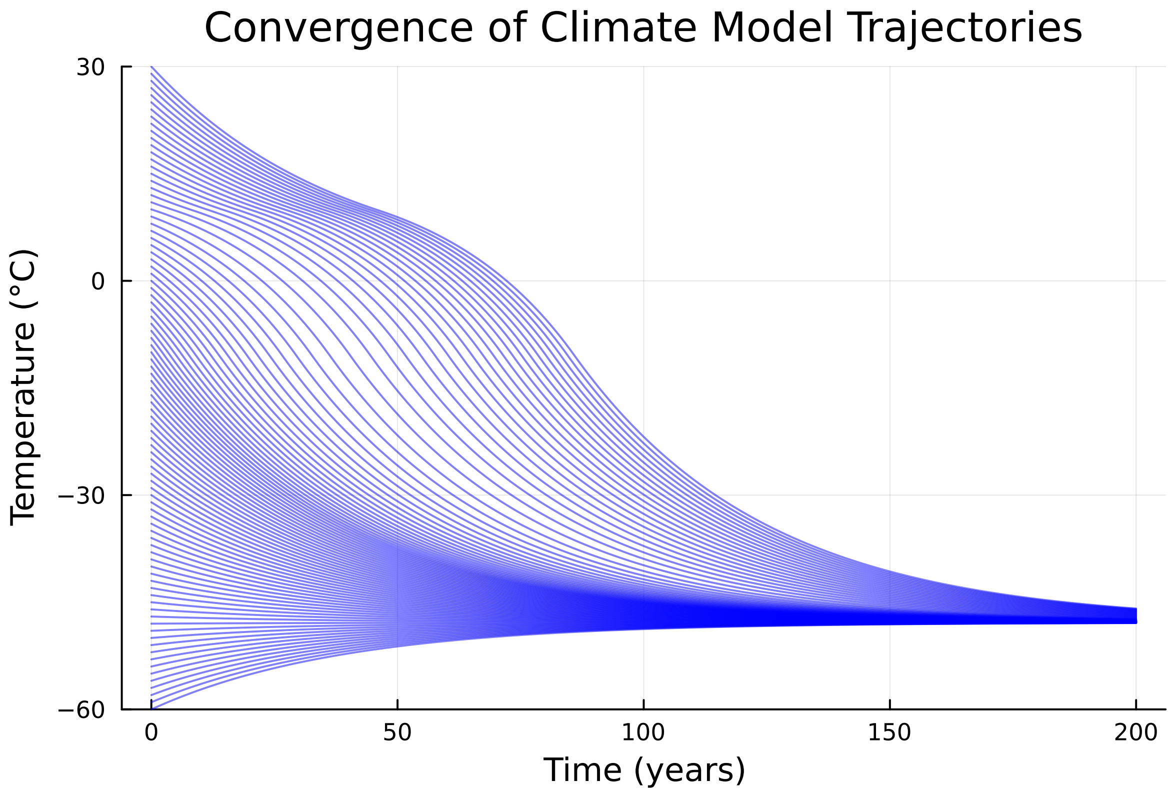
Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
C  = 51       # heat capacity
S  = 1272    # solar constant (W/m^2) - Neoprotereozoic value
A  = 221.2      # outgoing longwave radiation at T=0
B  = -1.3        # OLR temperature slope

#albedo
ai = 0.5
a0 = 0.3

dt = 1.0
years = 200
initial_T = -60:1:30   # 91 initial conditions
time = 0:years

#make matrix, length of the number of initial temperatures (91 of them from -60:30),
#width of number of years (0:200)
trajectories = Matrix{Float64}(undef, length(initial_T), years+1)

#go through all initial T's one by one
```

```julia
for (idx, T0) in enumerate(initial_T)
    #place T arrray for an initial T into the years array
    T_vals = zeros(years+1)
    T_vals[1] = T0

    for i in 1:years
        Ti = T_vals[i]

        # pick albedo value
        if Ti <= -10
            at = ai
        elseif -10 <= Ti <= 10
            at = ai + (a0 - ai) * ((Ti + 10) / 20)
        else
            at = a0
        end

        T_vals[i+1] = Ti + (dt / C) * (((1 - at) * S) / 4 - (A - B*Ti))
    end

    trajectories[idx, :] .= T_vals
end

plot(time, trajectories',
     xlabel="Time (years)",
     ylabel="Temperature (°C)",
     title="Convergence of Climate Model Trajectories",
     legend=false,
     color=:blue,
     alpha=0.5,
     ylim=(-60,30))
```



Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

References

List any external references consulted, including classmates.